

questions and 75.35% IK-IDK questions, which means Natural Questions is more challenging than TriviaQA. The results on Natural Questions are similar to those on TriviaQA. The aligned models show improvements in all metrics compared to using prompts. In contrast to the results on TriviaQA, Idk-HIR achieves the highest TRUTHFUL rate, rather than Idk-BoN. Furthermore, the models aligned using preference optimization methods exhibit a reduction in the TRUTHFUL rate compared to the Idk-SFT. We believe this is due to the fact that preference optimization encourages the model to answer more questions. We can observe that, compared to the Idk-SFT model, preference-optimized models have more IK-IK questions but less IK-IDK questions. In addition to this, we utilize ALCUNA to construct the Idk dataset, which only contains ID-IDK questions. The results from Table 1 indicate that the prompting method can already enable the model to refuse answering most unanswerable questions. After alignment, the model achieves an even higher IK-IDK rate. The model aligned with TriviaQA demonstrates a high TRUTHFUL rate on Natural Questions and a high IK-IDK rate on ALCUNA, suggesting that the model’s behavior of refusing to answer unknown questions can be generalized to OOD data.

4.4 Ablation Study

In this section, we analyze which factors affect the model’s ability to recognize questions it does not know through extensive ablation experiments.

	IK-IK↑	IK-IDK↑	IDK-IK↓	IDK-IDK↓	TRUTHFUL↑
Idk-SFT _{7b}	28.57	46.19	19.24	6.00	74.75
<i>w/Llama-2-13b-chat</i>	33.92	41.43	17.45	7.20	75.35 ^{↑0.60}
<i>w/Llama-2-70b-chat</i>	57.78	22.68	10.78	8.66	80.55 ^{↑5.8}
<i>w/Idk-Mistral</i>	18.35	50.65	27.68	3.31	69.00 ^{↓5.75}
<i>w/Idk-Baichuan</i>	8.85	53.07	36.37	1.71	61.92 ^{↓12.83}

Table 2: Results of ablation experiments.

Effect of model size The capabilities of LLMs are often closely related to the number of their parameters: models with a larger size tend to be more powerful. We conduct Idk-SFT on Llama-2-7b-chat, Llama-2-13b-chat and Llama-2-70b-chat to observe whether the size of the model affects the effectiveness of Idk-SFT. In Table 2, we report the proportions of each knowledge quadrant for models of different sizes. However, the label distribution of the Idk dataset corresponding to different initial models is inconsistent (the larger the model, the more IK-IK questions), as shown in Appendix A.3. This results in the IK-IK rate and IK-IDK rate being incomparable. Therefore, we mainly focus on the TRUTHFUL rate of different models. The TRUTHFUL rate of the 13B model is slightly higher than that of the 7B model. The TRUTHFUL rate of the 70B model is significantly higher than that of the 13B and 7B models. This indicates that larger models are more adept at distinguishing between questions they know and do not know.

Effect of data sources Due to the differences in the pre-training process, different pre-trained models possess distinct knowledge. During training, we construct model-specific Idk dataset for different pre-trained models. This is because we want the model to determine whether it knows the answer to a question based on its internal knowledge, rather than learning to recognize some specific patterns of questions. The model-specific Idk dataset can connect the model’s internal knowledge with the labels of the Idk dataset. To explore the impact of using a non-model-specific Idk dataset on training, we construct two Idk training sets using Mistral-7B-Instruct-v0.1 (Jiang et al., 2023) and Baichuan2-7B-chat (Baichuan, 2023) respectively, named “Idk-Mistral” and “Idk-Baichuan”. We present label distributions of these Idk datasets in Appendix A.3 As shown in Tabel 2, using non-model-specific Idk datasets like “Idk-Mistral” or “Idk-Baichuan” does result in a TRUTHFUL rate loss. Due to the numerous Idk questions in the Idk-Mistral and Idk-Baichuan datasets, the trained model tends to be more inclined towards refusing to answer questions, which has resulted

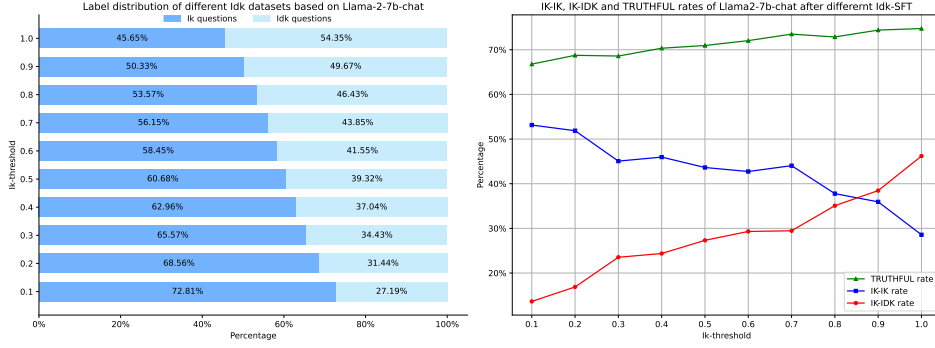


Figure 4: **Left:** Variation in the proportions of Ik and Idk questions within the Idk datasets constructed based on different Ik thresholds. **Right:** The changes in IK-IK rate, IK-IDK rate, and TRUTHFUL rate after conducting Idk-SFT with different Idk datasets.

in a significant reduction in Ik-Ik related queries, far below their proportion in the dataset. This indicates that constructing a model-specific Idk dataset is necessary for enabling the model to learn to refuse to answer questions it does not know.

Effect of IK threshold So far, we have fixed the Ik threshold to 1.0. Here, we discuss the impact of different Ik thresholds on model behaviors. The Ik threshold primarily affects the distribution of labels in the Idk dataset, with a higher Ik threshold indicating that more questions will be labeled as “I don’t know”. As demonstrated in Figure 4 (left), the higher the value of the Ik threshold, the greater the proportion of Idk questions. This is because when the Ik threshold is high, only questions with a high confidence level will be annotated as questions known to the model. As shown in Figure 4 (right), increasing the Ik threshold results in a decrease in the IK-IK rate and an increase in the IK-IDK rate. As the Ik threshold is raised, the model’s TRUTHFUL rate will continue to improve. In other words, setting a high Ik threshold aids the model in better distinguish between knowledge it knows and does not know, making the model more truthful. In contrast, setting a low Ik threshold can make the model more helpful, since the number of IK-IK questions will increase. Besides, we find that as the proportion of Idk questions in the dataset increases, the model tends to refuse to answer questions more frequently. We report the F1 scores of Idk and Ik questions in different Idk datasets in Appendix C.2 and the knowledge quadrants under different Ik thresholds in C.1.

5 Conclusion

In this paper, we explore the question “Can AI assistants know what they don’t know?”. We find that after aligning the AI assistant with its own Idk(“I don’t know”) dataset which contains its known and unknown questions, the assistant can be aware of what it does not know to a certain extent. In the given test set for open-domain question-answering, Llama-2-7b-chat can explicitly indicate whether it knows or does not know the answers to up to 78.96% of the questions and refuse to answer the questions it does not know. To achieve this, we utilize various methods to use the Idk dataset for alignment, including prompting, supervised fine-tuning and preference-aware optimization. We also find that the Ik threshold influences the model’s tendency to decline responses. Empolying an Idk dataset from different models for alignment results in a performance degradation. Furthermore, a large model like Llama-2-70b-chat achieves a higher TRUTHFUL rate. An AI assistant’s refusal to answer questions beyond its knowledge can reduce hallucinations. We believe this is an essential behavior for a truthful AI assistant.

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